MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 11

1. *Let* $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$ *and* $\mathbf{b} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$.

(1) Does $\mathbf{b}$ belong to the column space of A? Can you solve $A\mathbf{x} = \mathbf{b}$?

No, $\mathbf{b}$ does not belong to the column space of A, because it is not a linear combination of $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$. Hence there is no solution to $A\mathbf{x} = \mathbf{b}$.

(2) What do you expect the projection of $\mathbf{b}$ onto $W = \text{Col}(A)$ to be?

W is the span of $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$. It can easily be seen that W is the set of vectors in $\mathbb{R}^3$ whose third entry is 0. Hence $\begin{bmatrix} 4 \\ 5 \\ 0 \end{bmatrix}$ is in W. Note

$$\begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix} = \mathbf{b} - \begin{bmatrix} 4 \\ 5 \\ 0 \end{bmatrix}$$

is orthogonal to W. Hence $\begin{bmatrix} 4 \\ 5 \\ 0 \end{bmatrix}$ should be the orthogonal projection of $\mathbf{b}$ onto W.

(3) Find the projection $\hat{\mathbf{b}}$ of $\mathbf{b}$ onto $\text{Col}(A)$, and then solve $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$. (The vector $\hat{\mathbf{x}}$ is called *the least square solution* of $A\mathbf{x} = \mathbf{b}$.)

Step 1: Find the orthogonal projection of $\mathbf{b}$ onto W.

$$\hat{\mathbf{b}} = \frac{\begin{bmatrix}4\\5\\6\end{bmatrix} \cdot \begin{bmatrix}1\\1\\0\end{bmatrix}}{\begin{bmatrix}1\\1\\0\end{bmatrix} \cdot \begin{bmatrix}1\\1\\0\end{bmatrix}} \begin{bmatrix}1\\1\\0\end{bmatrix} + \frac{\begin{bmatrix}4\\5\\6\end{bmatrix} \cdot \begin{bmatrix}-1\\1\\0\end{bmatrix}}{\begin{bmatrix}-1\\1\\0\end{bmatrix} \cdot \begin{bmatrix}-1\\1\\0\end{bmatrix}} \begin{bmatrix}-1\\1\\0\end{bmatrix} = \begin{bmatrix}4.5\\4.5\\0\end{bmatrix} + \begin{bmatrix}-.5\\.5\\0\end{bmatrix} = \begin{bmatrix}4\\5\\0\end{bmatrix}.$$

Step 2: Solve $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$.

$$\left[\begin{array}{cc|c} 1 & -1 & 4 \\ 1 & 1 & 5 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow{R2 \to R2 - R_1} \left[\begin{array}{cc|c} 1 & -1 & 4 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow{R1 \to R1 + .5R2} \left[\begin{array}{cc|c} 1 & 0 & 4.5 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{array}\right].$$

Hence,

$$\hat{\mathbf{x}} = \begin{bmatrix}4.5\\0.5\end{bmatrix}.$$

(4) Solve the equation $A^TA\hat{\mathbf{x}} = A^T\mathbf{b}$. Compare with your result of the previous part! (This equation is called the *normal equation* of $A\mathbf{x} = \mathbf{b}$.)

We first calculate A^TA and $A^T\mathbf{b}$:

$$A^TA = \begin{bmatrix}1 & 1 & 0\\-1 & 1 & 0\end{bmatrix} \begin{bmatrix}1 & -1\\1 & 1\\0 & 0\end{bmatrix} = \begin{bmatrix}2 & 0\\0 & 2\end{bmatrix},$$

$$A^T\mathbf{b} = \begin{bmatrix}1 & 1 & 0\\-1 & 1 & 0\end{bmatrix} \begin{bmatrix}4\\5\\6\end{bmatrix} = \begin{bmatrix}9\\1\end{bmatrix}.$$

Now we have to solve

$$\begin{bmatrix}2 & 0\\0 & 2\end{bmatrix} \hat{\mathbf{x}} = \begin{bmatrix}9\\1\end{bmatrix}.$$

It is easy to check that then

$$\hat{\mathbf{x}} = \begin{bmatrix}4.5\\0.5\end{bmatrix}.$$

2. *True or false? If true, give a brief explanation why. If false, give a counterexample. All vectors below are in $\mathbb{R}^n$ for some n.*

(1) Let W be a subspace of $\mathbb{R}^n$ and $\mathbf{a}, \mathbf{b}, \mathbf{c}$ in $\mathbb{R}^n$ such that $\mathbf{a} = \mathbf{b} + \mathbf{c}$ and $\mathbf{b}$ is in W and $\mathbf{c}$ is $W^\perp$. Then $\mathbf{b}$ is the projection of $\mathbf{a}$ onto W.

True. Observe that $\mathbf{c} = \mathbf{a} - \mathbf{b}$ is in $W^\perp$. Since the projection of $\mathbf{a}$ onto W is the unique vector $\mathbf{v}$ in W such that $\mathbf{a} - \mathbf{v}$ is in $W^\perp$, we get that $\mathbf{b}$ is the projection of $\mathbf{a}$ onto W.

(2) All projection matrices are invertible.

False. If $W = Span\{\begin{bmatrix}1\\0\end{bmatrix}\}$, then $P = \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}$

(3) If $\mathbf{w}$ is a vector in W and T denotes the projection onto the subspace W, then $T(\mathbf{w}) = \mathbf{w}$.

True: Note that $\mathbf{w} = \underbrace{\mathbf{w}}_{\in W} + \underbrace{\mathbf{0}}_{\in W^\perp}$. Thus by (1), $\mathbf{w}$ is the projection of $\mathbf{w}$ onto W.

(4) If $\mathbf{v}$ is a vector in $W^\perp$ and T denotes the projection onto the subspace W, then $T(\mathbf{v}) = 0$.

Note that $\mathbf{w} = \underbrace{\mathbf{0}}_{\in W} + \underbrace{\mathbf{w}}_{\in W^\perp}$. Thus by (1), 0 is the projection of $\mathbf{w}$ onto W.

(5) If P be the projection matrix of a projection onto a subspace W, then $\text{Nul}(P) = W^\perp$.

True. By the previous question, for every $\mathbf{v}$ in $W^\perp$, we have $P\mathbf{v} = \mathbf{0}$. Now suppose that $\mathbf{v}$ is the null space of P. Thus $P\mathbf{v} = 0$. Since $\mathbf{v} - P\mathbf{v}$ is in $W^\perp$, we get that $\mathbf{v} - P\mathbf{v} = \mathbf{v}$ is in $W^\perp$.

(6) If $W = \mathbb{R}^n$, that is, we consider W to be the whole vector space $\mathbb{R}^n$, then the projection matrix (in the standard basis) is the identity matrix.

True, because this projection maps the standard basis to itself.

3. *Let A be a matrix with linearly independent columns. From class, we know that the projection matrix P of the projection onto the column space of A is $A(A^TA)^{-1}A^T$.*

(1) What is the projection matrix of the orthogonal projection onto the row space of A?

Just replace A by A^T and obtain $A^T(AA^T)^{-1}A$.

(2) What is the projection matrix for the orthogonal projection onto the left null space of A?

Note that the left null space is the orthogonal complement of the column space of A. If Q is the projection matrix onto the left null space and P is projection matrix onto the column space, then $Q = I - P$, where is the identity matrix.

(3) Why is P symmetric (ie. $P^T = P$)?

Note that

$$(A(A^TA)^{-1}A^T)^T = (A^T)^T((A^TA)^{-1})^TA^T = A(A^T(A^T)^T)^{-1}A^T = A(A^TA)^{-1}A^T$$

4. *Let* $W = \text{span}\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\-1\\0\end{bmatrix}\right\}$ *and* $V = \text{span}\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\1\\-2\end{bmatrix}\right\}$ *be subspaces of* $\mathbb{R}^3$.

(a) Find the projection matrices, P and Q, corresponding to the projections onto W and V, respectively.

$$\begin{bmatrix}1\\0\\0\end{bmatrix}_W = \frac{\begin{bmatrix}1\\0\\0\end{bmatrix}^T\begin{bmatrix}1\\1\\1\end{bmatrix}}{\begin{bmatrix}1\\1\\1\end{bmatrix}^T\begin{bmatrix}1\\1\\1\end{bmatrix}}\begin{bmatrix}1\\1\\1\end{bmatrix} + \frac{\begin{bmatrix}1\\0\\0\end{bmatrix}^T\begin{bmatrix}1\\-1\\0\end{bmatrix}}{\begin{bmatrix}1\\-1\\0\end{bmatrix}^T\begin{bmatrix}1\\-1\\0\end{bmatrix}}\begin{bmatrix}1\\-1\\0\end{bmatrix} = \frac{1}{3}\begin{bmatrix}1\\1\\1\end{bmatrix} + \frac{1}{2}\begin{bmatrix}1\\-1\\0\end{bmatrix} = \begin{bmatrix}\frac{5}{6}\\-\frac{1}{6}\\\frac{1}{3}\end{bmatrix},$$

$$\begin{bmatrix}0\\1\\0\end{bmatrix}_W = \frac{\begin{bmatrix}0\\1\\0\end{bmatrix}^T\begin{bmatrix}1\\1\\1\end{bmatrix}}{\begin{bmatrix}1\\1\\1\end{bmatrix}^T\begin{bmatrix}1\\1\\1\end{bmatrix}}\begin{bmatrix}1\\1\\1\end{bmatrix} + \frac{\begin{bmatrix}0\\1\\0\end{bmatrix}^T\begin{bmatrix}1\\-1\\0\end{bmatrix}}{\begin{bmatrix}1\\-1\\0\end{bmatrix}^T\begin{bmatrix}1\\-1\\0\end{bmatrix}}\begin{bmatrix}1\\-1\\0\end{bmatrix} = \frac{1}{3}\begin{bmatrix}1\\1\\1\end{bmatrix} - \frac{1}{2}\begin{bmatrix}1\\-1\\0\end{bmatrix} = \begin{bmatrix}-\frac{1}{6}\\\frac{5}{6}\\\frac{1}{3}\end{bmatrix},$$

$$\begin{bmatrix}0\\0\\1\end{bmatrix}_W = \frac{\begin{bmatrix}0\\0\\1\end{bmatrix}^T\begin{bmatrix}1\\1\\1\end{bmatrix}}{\begin{bmatrix}1\\1\\1\end{bmatrix}^T\begin{bmatrix}1\\1\\1\end{bmatrix}}\begin{bmatrix}1\\1\\1\end{bmatrix} + \frac{\begin{bmatrix}0\\0\\1\end{bmatrix}^T\begin{bmatrix}1\\-1\\0\end{bmatrix}}{\begin{bmatrix}1\\-1\\0\end{bmatrix}^T\begin{bmatrix}1\\-1\\0\end{bmatrix}}\begin{bmatrix}1\\-1\\0\end{bmatrix} = \frac{1}{3}\begin{bmatrix}1\\1\\1\end{bmatrix} + 0\begin{bmatrix}1\\-1\\0\end{bmatrix} = \begin{bmatrix}\frac{1}{3}\\\frac{1}{3}\\\frac{1}{3}\end{bmatrix},$$

Hence, the projection matrix corresponding to the orthogonal projection onto W is:

$$P = \begin{bmatrix}\frac{5}{6} & -\frac{1}{6} & \frac{1}{3}\\-\frac{1}{6} & \frac{5}{6} & \frac{1}{3}\\\frac{1}{3} & \frac{1}{3} & \frac{1}{3}\end{bmatrix}$$

Also,

$$\begin{bmatrix}1\\0\\0\end{bmatrix}_V = \frac{\begin{bmatrix}1\\0\\0\end{bmatrix}^T\begin{bmatrix}1\\1\\1\end{bmatrix}}{\begin{bmatrix}1\\1\\1\end{bmatrix}^T\begin{bmatrix}1\\1\\1\end{bmatrix}}\begin{bmatrix}1\\1\\1\end{bmatrix} + \frac{\begin{bmatrix}1\\0\\0\end{bmatrix}^T\begin{bmatrix}1\\1\\-2\end{bmatrix}}{\begin{bmatrix}1\\1\\-2\end{bmatrix}^T\begin{bmatrix}1\\1\\-2\end{bmatrix}}\begin{bmatrix}1\\1\\-2\end{bmatrix} = \frac{1}{3}\begin{bmatrix}1\\1\\1\end{bmatrix} + \frac{1}{6}\begin{bmatrix}1\\1\\-2\end{bmatrix} = \begin{bmatrix}\frac{1}{2}\\\frac{1}{2}\\0\end{bmatrix},$$

$$\begin{bmatrix}0\\1\\0\end{bmatrix}_V = \frac{\begin{bmatrix}0\\1\\0\end{bmatrix}^T\begin{bmatrix}1\\1\\1\end{bmatrix}}{\begin{bmatrix}1\\1\\1\end{bmatrix}^T\begin{bmatrix}1\\1\\1\end{bmatrix}}\begin{bmatrix}1\\1\\1\end{bmatrix} + \frac{\begin{bmatrix}0\\1\\0\end{bmatrix}^T\begin{bmatrix}1\\1\\-2\end{bmatrix}}{\begin{bmatrix}1\\1\\-2\end{bmatrix}^T\begin{bmatrix}1\\1\\-2\end{bmatrix}}\begin{bmatrix}1\\1\\-2\end{bmatrix} = \frac{1}{3}\begin{bmatrix}1\\1\\1\end{bmatrix} + \frac{1}{6}\begin{bmatrix}1\\1\\-2\end{bmatrix} = \begin{bmatrix}\frac{1}{2}\\\frac{1}{2}\\0\end{bmatrix},$$

$$\begin{bmatrix}0\\0\\1\end{bmatrix}_V = \frac{\begin{bmatrix}0\\0\\1\end{bmatrix}^T\begin{bmatrix}1\\1\\1\end{bmatrix}}{\begin{bmatrix}1\\1\\1\end{bmatrix}^T\begin{bmatrix}1\\1\\1\end{bmatrix}}\begin{bmatrix}1\\1\\1\end{bmatrix} + \frac{\begin{bmatrix}0\\0\\1\end{bmatrix}^T\begin{bmatrix}1\\1\\-2\end{bmatrix}}{\begin{bmatrix}1\\1\\-2\end{bmatrix}^T\begin{bmatrix}1\\1\\-2\end{bmatrix}}\begin{bmatrix}1\\1\\-2\end{bmatrix} = \frac{1}{3}\begin{bmatrix}1\\1\\1\end{bmatrix} - \frac{1}{3}\begin{bmatrix}1\\1\\-2\end{bmatrix} = \begin{bmatrix}0\\0\\1\end{bmatrix},$$

Hence, the projection matrix corresponding to the orthogonal projection onto V is:

$$Q = \begin{bmatrix}\frac{1}{2} & \frac{1}{2} & 0\\ \frac{1}{2} & \frac{1}{2} & 0\\ 0 & 0 & 1\end{bmatrix}$$

(b) Check that $PQ = QP$. Can you interpret PQ as a projection matrix?

$PQ = QP$ is the matrix corresponding to the orthogonal projection onto $W \cap V = \text{span}\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}\right\}$.

[Note: since $\begin{bmatrix}1\\-1\\0\end{bmatrix}^T\begin{bmatrix}1\\1\\-2\end{bmatrix} = 0$ if you compute orthogonal projection onto W and then onto V the answer will be same as computing orthogonal projection onto V and then onto W]

(c) Compute the matrices $P_{\mathcal{BB}}$ and $Q_{\mathcal{BB}}$ for projection onto W and B with respect to the input and output basis:

$$\mathcal{B} = \left\{\begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\-1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\-1\end{bmatrix}\right\}$$

We first compute $P_{\mathcal{BB}}$ by projecting the vectors from $\mathcal{B}$ onto W and then converting them to $\mathcal{B}$-coordinates:

$$\pi_W\left(\begin{bmatrix}1\\1\\1\end{bmatrix}\right) = \begin{bmatrix}1\\1\\1\end{bmatrix} = 1\begin{bmatrix}1\\1\\1\end{bmatrix} + 0\begin{bmatrix}1\\-1\\0\end{bmatrix} + 0\begin{bmatrix}1\\1\\-1\end{bmatrix} \implies \pi_W\left(\begin{bmatrix}1\\1\\1\end{bmatrix}\right)_{\mathcal{B}} = \begin{bmatrix}1\\0\\0\end{bmatrix}$$

$$\pi_W\left(\begin{bmatrix}1\\-1\\0\end{bmatrix}\right) = \begin{bmatrix}1\\-1\\0\end{bmatrix} = 0\begin{bmatrix}1\\1\\1\end{bmatrix} + 1\begin{bmatrix}1\\-1\\0\end{bmatrix} + 0\begin{bmatrix}1\\1\\-1\end{bmatrix} \implies \pi_W\left(\begin{bmatrix}1\\-1\\0\end{bmatrix}\right)_{\mathcal{B}} = \begin{bmatrix}0\\1\\0\end{bmatrix}$$

$$\pi_W\left(\begin{bmatrix}1\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}5/6 & -1/6 & 1/3\\-1/6 & 5/6 & 1/3\\1/3 & 1/3 & 1/3\end{bmatrix}\begin{bmatrix}1\\1\\-1\end{bmatrix} = \begin{bmatrix}1/3\\1/3\\1/3\end{bmatrix} = \frac{1}{3}\begin{bmatrix}1\\1\\1\end{bmatrix} + 0\begin{bmatrix}1\\-1\\0\end{bmatrix} + 0\begin{bmatrix}1\\1\\-1\end{bmatrix}$$

$$\implies \pi_W\left(\begin{bmatrix}1\\1\\-1\end{bmatrix}\right)_{\mathcal{B}} = \begin{bmatrix}\frac{1}{3}\\0\\0\end{bmatrix}$$

We conclude that

$$P_{\mathcal{BB}} = \begin{bmatrix}1 & 0 & \frac{1}{3}\\0 & 1 & 0\\0 & 0 & 0\end{bmatrix}.$$

Next we do the same thing for V:

$$\pi_V\left(\begin{bmatrix}1\\1\\1\end{bmatrix}\right) = \begin{bmatrix}1\\1\\1\end{bmatrix} = 1\begin{bmatrix}1\\1\\1\end{bmatrix} + 0\begin{bmatrix}1\\-1\\0\end{bmatrix} + 0\begin{bmatrix}1\\1\\-1\end{bmatrix} \implies \pi_V\left(\begin{bmatrix}1\\1\\1\end{bmatrix}\right)_{\mathcal{B}} = \begin{bmatrix}1\\0\\0\end{bmatrix}$$

$$\pi_V\left(\begin{bmatrix}1\\-1\\0\end{bmatrix}\right) = \begin{bmatrix}1/2 & 1/2 & 0\\1/2 & 1/2 & 0\\0 & 0 & 1\end{bmatrix}\begin{bmatrix}1\\-1\\0\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix} = 0\begin{bmatrix}1\\1\\1\end{bmatrix} + 0\begin{bmatrix}1\\-1\\0\end{bmatrix} + 0\begin{bmatrix}1\\1\\-1\end{bmatrix}$$

$$\implies \pi_V\left(\begin{bmatrix}1\\-1\\0\end{bmatrix}\right)_{\mathcal{B}} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

$$\pi_V\left(\begin{bmatrix}1\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1/2 & 1/2 & 0\\1/2 & 1/2 & 0\\0 & 0 & 1\end{bmatrix}\begin{bmatrix}1\\1\\-1\end{bmatrix} = \begin{bmatrix}1\\1\\-1\end{bmatrix} = 0\begin{bmatrix}1\\1\\1\end{bmatrix} + 0\begin{bmatrix}1\\-1\\0\end{bmatrix} + 1\begin{bmatrix}1\\1\\-1\end{bmatrix}$$

$$\implies \pi_V\left(\begin{bmatrix}1\\1\\-1\end{bmatrix}\right)_{\mathcal{B}} = \begin{bmatrix}0\\0\\1\end{bmatrix}.$$

We conclude that

$$Q_{\mathcal{BB}} = \begin{bmatrix}1 & 0 & 0\\0 & 0 & 0\\0 & 0 & 1\end{bmatrix}$$

(d) What is the product of $P_{\mathcal{BB}}$ and $Q_{\mathcal{BB}}$? How does your answer relate to the answer in part (b)?

We compute

$$P_{\mathcal{BB}}Q_{\mathcal{BB}} = \begin{bmatrix} 1 & 0 & \frac{1}{3} \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & \frac{1}{3} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Again, this is the projection matrix onto the subspace spanned by $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, but with respect to the basis $\mathcal{B}$.

5. *Recall that the orthogonal projection onto* $\mathrm{Col}(A)$ *has projection matrix* $A(A^TA)^{-1}A^T$.

(1) How does this formula simplify in the case when A has orthonormal columns?

If A has orthonormal columns then $A^TA = I$. So the projection matrix is:

$$A(A^TA)^{-1}A^T = AA^T$$

(2) Let $Q = \begin{bmatrix} 1 & 0 \\ 0 & \frac{3}{5} \\ 0 & -\frac{4}{5} \end{bmatrix}$. What is the projection matrix corresponding to the orthogonal projection onto $\mathrm{Col}(Q)$?

Since Q has orthonormal columns, the projection matrix is:

$$QQ^T = \begin{bmatrix} 1 & 0 \\ 0 & \frac{3}{5} \\ 0 & -\frac{4}{5} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & -\frac{4}{5} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{9}{25} & -\frac{12}{25} \\ 0 & -\frac{12}{25} & \frac{16}{25} \end{bmatrix}$$

(3) Let $Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & \frac{4}{5} \\ 0 & -\frac{4}{5} & \frac{3}{5} \end{bmatrix}$. What is the projection matrix corresponding to the orthogonal projection onto $\mathrm{Col}(Q)$? Explain why your answer is not surprising.

Q has orthonormal columns, so the projection matrix is QQ^T which is equal to I(since Q is orthogonal).Since columns of Q are linearly independent and Q has 3 columns, columns of Q form a basis for $\mathbb{R}^3$. Hence, $\mathrm{Col}(Q) = \mathbb{R}^3$ and projection of each vector in $\mathbb{R}^3$ onto $\mathrm{Col}(Q)$ is itself, i.e., the projection matrix is I.

6. *Let* A *be an* $m \times n$*-matrix*

(1) Explain why $\mathrm{Nul}(A) = \mathrm{Nul}(A^TA)$.

If $\mathbf{z} \in \mathrm{Nul}(A)$, then $A^TA\mathbf{z} = A^T\mathbf{0} = \mathbf{0}$. Thus $\mathbf{z} \in \mathrm{Nul}(A^TA)$. Now suppose that $\mathbf{z}$ is in $\mathrm{Nul}(A^TA)$. Then $A^TA\mathbf{z} = \mathbf{0}$. However, then

$$(A\mathbf{z}) \cdot (A\mathbf{z}) = (A\mathbf{z})^T(A\mathbf{z}) = \mathbf{z}^TA^TA\mathbf{z} = \mathbf{z}^T\mathbf{0} = 0.$$

Thus $A\mathbf{z} = \mathbf{0}$, since the zero vector is the only vector whose dot product with itself is 0.

(2) Suppose A has linearly independent columns. Why is A^TA invertible?

Note that A^TA is an $n \times n$ matrix. Since A has linearly independent columns, $\mathrm{Nul}(A) = \{\mathbf{0}\}$. Thus by (1), $\mathrm{Nul}(A^TA) = \{\mathbf{0}\}$. There A^TA is invertible.

7. (a) Given the following orthogonal set of vectors in $\mathbb{R}^4$:

$$\left\{ \begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ -3 \\ 0 \end{bmatrix} \right\}$$

use the FTLA (fundamental theorem of linear algebra) to complete the set to an orthogonal basis $\mathcal{B}$ for $\mathbb{R}^4$ (i.e., find a 4th vector to add to the set so that the set of 4 vectors is an orthogonal basis of $\mathbb{R}^4$). Hint: write the span, V, of this set as the column space of some matrix A, then find a basis for $V^\perp$.

Consider the following matrix A:

$$A = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 1 & 0 \\ 2 & 0 & -3 \\ 0 & 1 & 0 \end{bmatrix}$$

It is clear that $V = \text{Col}(A)$. We want to find a basis for $V^\perp$. By FTLA, this is equivalent to finding a basis for $\text{Nul}(A^T)$ (which conveniently is something we can easily compute). So we row reduce...

$$\begin{bmatrix} 2 & 0 & 2 & 0 \\ 0 & 1 & 0 & 1 \\ 3 & 0 & -3 & 0 \end{bmatrix} \xrightarrow{R_1 \to \frac{1}{2}R_1} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 3 & 0 & -3 & 0 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 3R_1} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -6 & 0 \end{bmatrix}$$

$$\xrightarrow{R_3 \to -\frac{1}{6}R_3} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow{R_1 \to R_1 - R_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

We identify x_4 as a free variable, x_1, x_2 and x_3 as pivot variables, and we write down the general solution to the homogeneous equation:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ -x_4 \\ 0 \\ x_4 \end{bmatrix} = x_4 \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

We conclude that

$$\text{Nul}(A^T) = V^\perp = \text{span} \left\{ \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

This allows us to complete the original set of 3 orthogonal vectors to an orthogonal basis of $\mathbb{R}^4$:

$$\left\{ \begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ -3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

(b) Let V be the span of the orthogonal set in part (a). Let $\pi_V : \mathbb{R}^4 \to \mathbb{R}^4$ be the projection onto V. Find the coordinate matrix $P_{\mathcal{B},\mathcal{B}}$ for π_V using $\mathcal{B}$ from part (a) as the input and output basis.

We do this the way we would represent any linear transformation given bases for the input and output vector space:

$$\begin{bmatrix}2\\0\\2\\0\end{bmatrix} = \pi_V\left(\begin{bmatrix}2\\0\\2\\0\end{bmatrix}\right) = 1\begin{bmatrix}2\\0\\2\\0\end{bmatrix} + 0\begin{bmatrix}0\\1\\0\\1\end{bmatrix} + 0\begin{bmatrix}3\\0\\-3\\0\end{bmatrix} + 0\begin{bmatrix}0\\-1\\0\\1\end{bmatrix}$$

$$\begin{bmatrix}0\\1\\0\\1\end{bmatrix} = \pi_V\left(\begin{bmatrix}0\\1\\0\\1\end{bmatrix}\right) = 0\begin{bmatrix}2\\0\\2\\0\end{bmatrix} + 1\begin{bmatrix}0\\1\\0\\1\end{bmatrix} + 0\begin{bmatrix}3\\0\\-3\\0\end{bmatrix} + 0\begin{bmatrix}0\\-1\\0\\1\end{bmatrix}$$

$$\begin{bmatrix}3\\0\\-3\\0\end{bmatrix} = \pi_V\left(\begin{bmatrix}3\\0\\-3\\0\end{bmatrix}\right) = 0\begin{bmatrix}2\\0\\2\\0\end{bmatrix} + 0\begin{bmatrix}0\\1\\0\\1\end{bmatrix} + 1\begin{bmatrix}3\\0\\-3\\0\end{bmatrix} + 0\begin{bmatrix}0\\-1\\0\\1\end{bmatrix}$$

$$\begin{bmatrix}0\\0\\0\\0\end{bmatrix} = \pi_V\left(\begin{bmatrix}0\\-1\\0\\1\end{bmatrix}\right) = 0\begin{bmatrix}2\\0\\2\\0\end{bmatrix} + 0\begin{bmatrix}0\\1\\0\\1\end{bmatrix} + 0\begin{bmatrix}3\\0\\-3\\0\end{bmatrix} + 0\begin{bmatrix}0\\-1\\0\\1\end{bmatrix}$$

and so

$$P_{\mathcal{B},\mathcal{B}} = \begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&0\end{bmatrix}$$

where for the last column of the matrix we use the fact that the vector $\begin{bmatrix}0\\-1\\0\\1\end{bmatrix}$ is in $V^\perp$ so its projection onto V is $\begin{bmatrix}0\\0\\0\\0\end{bmatrix}$.

(c) Let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be an orthogonal set of vectors in $\mathbb{R}^5$. Let $V = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ and let $\mathcal{C} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$ be an orthogonal basis for $\mathbb{R}^5$ (so $\mathcal{C}$ is the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ together with two additional vectors $\mathbf{v}_4$ and $\mathbf{v}_5$ which "completes it" to an orthogonal basis of $\mathbb{R}^5$). If $\pi_V : \mathbb{R}^5 \to \mathbb{R}^5$ is projection onto V, what is the matrix $P_{\mathcal{C},\mathcal{C}}$ for π_V with respect to input and output basis $\mathcal{C}$?

We need to do essentially the same type of calculation that we did in part (b):

$$\begin{aligned}
\mathbf{v}_1 = \pi_V(\mathbf{v}_1) &= 1\mathbf{v}_1 + 0\mathbf{v}_2 + 0\mathbf{v}_3 + 0\mathbf{v}_4 + 0\mathbf{v}_5 \\
\mathbf{v}_2 = \pi_V(\mathbf{v}_2) &= 0\mathbf{v}_1 + 1\mathbf{v}_2 + 0\mathbf{v}_3 + 0\mathbf{v}_4 + 0\mathbf{v}_5 \\
\mathbf{v}_3 = \pi_V(\mathbf{v}_3) &= 0\mathbf{v}_1 + 0\mathbf{v}_2 + 1\mathbf{v}_3 + 0\mathbf{v}_4 + 0\mathbf{v}_5 \\
\mathbf{0} = \pi_V(\mathbf{v}_4) &= 0\mathbf{v}_1 + 0\mathbf{v}_2 + 0\mathbf{v}_3 + 0\mathbf{v}_4 + 0\mathbf{v}_5 \\
\mathbf{0} = \pi_V(\mathbf{v}_5) &= 0\mathbf{v}_1 + 0\mathbf{v}_2 + 0\mathbf{v}_3 + 0\mathbf{v}_4 + 0\mathbf{v}_5
\end{aligned}$$

where we use the fact that $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in V$ and $\mathbf{v}_4, \mathbf{v}_5 \in V^\perp$ to compute the values of π_V. Thus the matrix representation is:

$$P_{\mathcal{C},\mathcal{C}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

8. *A scientist tries to find the relation between the mysterious quantities x and y. She measures the following values:*

x	1	2	3	4
y	2	5	9	17

(i) Suppose that y is a linear function of x of the form $a + bx$. Set up the system of equations to find the coefficients a and b.

We set up the equation as follows:

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 9 \\ 17 \end{bmatrix}.$$

(ii) Find the best estimate for the coefficients a and b.

Use the least square solution. We calculate

$$\begin{bmatrix}1&1\\1&2\\1&3\\1&4\end{bmatrix}^T\begin{bmatrix}1&1\\1&2\\1&3\\1&4\end{bmatrix}=\begin{bmatrix}1&1&1&1\\1&2&3&4\end{bmatrix}\begin{bmatrix}1&1\\1&2\\1&3\\1&4\end{bmatrix}=\begin{bmatrix}4&10\\10&30\end{bmatrix}$$

and

$$\begin{bmatrix}1&1\\1&2\\1&3\\1&4\end{bmatrix}^T\begin{bmatrix}2\\5\\9\\17\end{bmatrix}=\begin{bmatrix}1&1&1&1\\1&2&3&4\end{bmatrix}\begin{bmatrix}2\\5\\9\\17\end{bmatrix}=\begin{bmatrix}33\\107\end{bmatrix}$$

Now we solve

$$\left[\begin{array}{cc|c}4&10&33\\10&30&107\end{array}\right]\xrightarrow{R2\to R2-2.5R_1}\left[\begin{array}{cc|c}4&10&33\\0&5&24.5\end{array}\right]\xrightarrow{R1\to R1-2R2}\left[\begin{array}{cc|c}4&0&-16\\0&5&24.5\end{array}\right].$$

Hence $a=-4$ and $b=4.9$.

(iii) Same question if we suppose that y is a quadratic function of the $a+bx+cx^2$.

We set up the equation as follows:

$$\begin{bmatrix}1&1&1\\1&2&4\\1&3&9\\1&4&16\end{bmatrix}\begin{bmatrix}a\\b\\c\end{bmatrix}=\begin{bmatrix}2\\5\\9\\17\end{bmatrix}.$$

We calculate

$$\begin{bmatrix}1&1&1\\1&2&4\\1&3&9\\1&4&16\end{bmatrix}^T\begin{bmatrix}1&1&1\\1&2&4\\1&3&9\\1&4&16\end{bmatrix}=\begin{bmatrix}1&1&1&1\\1&2&3&4\\1&4&9&16\end{bmatrix}\begin{bmatrix}1&1&1\\1&2&4\\1&3&9\\1&4&16\end{bmatrix}=\begin{bmatrix}4&10&30\\10&30&100\\30&100&354\end{bmatrix}$$

and

$$\begin{bmatrix}1&1&1\\1&2&4\\1&3&9\\1&4&16\end{bmatrix}^T\begin{bmatrix}2\\5\\9\\17\end{bmatrix}=\begin{bmatrix}1&1&1&1\\1&2&3&4\\1&4&9&16\end{bmatrix}\begin{bmatrix}2\\5\\9\\17\end{bmatrix}=\begin{bmatrix}33\\107\\375\end{bmatrix}.$$

One can row reduce

$$\left[\begin{array}{ccc|c}4&10&30&33\\10&30&100&107\\30&100&354&375\end{array}\right]\to\left[\begin{array}{ccc|c}1&0&0&2.25\\0&1&0&-1.35\\0&0&1&1.25\end{array}\right].$$

So $a=2.25$, $b=-1.35$ and $c=1.25$.

9. *Let A be an $m\times n$-matrix and let $\mathbf{b}\in\mathbb{R}^m$.*

(1) For which $\mathbf{b}$ is the following statement true:
If $\hat{\mathbf{x}}_1,\hat{\mathbf{x}}_2$ are least-squares solutions to $A\mathbf{x}=\mathbf{b}$, then $\hat{\mathbf{x}}_1+\hat{\mathbf{x}}_2$ is a least-squares solution to $A\mathbf{x}=\mathbf{b}$.

Only if $A^T\mathbf{b}$ is the zero vector. Observe that $\hat{\mathbf{x}}$ is a least-squares solution to $A\mathbf{x} = \mathbf{b}$ if and only if it is a solution to the linear system $A^TA\mathbf{x} = A^T\mathbf{b}$. However, if $\hat{\mathbf{x}_1}, \hat{\mathbf{x}_2}$ are solutions of $A^TA\mathbf{x} = A^T\mathbf{b}$, then their sum is only a solution as well if $A^T\mathbf{b}$ is the zero vector.

(2) Let $\hat{\mathbf{x}}$ be a least-squares solution to $A\mathbf{x} = \mathbf{b}$ and $\mathbf{y}$ in $\text{Nul}(A)$. Explain why $\hat{\mathbf{x}} + \mathbf{y}$ is a least-squares solution to $A\mathbf{x} = \mathbf{b}$.

Since $\hat{\mathbf{x}}$ is a least-squares solution, we have $A^TA\hat{\mathbf{x}} = A^T\mathbf{b}$. Thus

$$A^TA(\hat{\mathbf{x}} + y) = A^TA\hat{\mathbf{x}} + A^TAy = A^TA\hat{\mathbf{x}} + A^T\mathbf{0} = A^TA\hat{\mathbf{x}} = A^T\mathbf{b}.$$

Thus $\hat{\mathbf{x}} + y$ is a least-squares solution as well.

10. *Quarterly economic data is subject to seasonal fluctuations. A curve that approximates the gross domestic product (GDP) of a country might be of the form*

$$y = \beta_0 + \beta_1 x + \beta_2 \sin(2\pi x/4),$$

where is x is the time in quarters of a year. The term $\beta_0 + \beta_1 x$ gives the basic GDP growth trend of the economy, while the sine term reflects the seasonal changes. Assume the GDP data are $(x_1, y_1), \ldots, (x_n, y_n)$.

(i) Give the design matrix that leads to a least-square fit to the equation above.

Plugging in our data points we get that our design matrix is

$$\begin{bmatrix} 1 & x_1 & \sin(2\pi x_1/4) \\ 1 & x_2 & \sin(2\pi x_2/4) \\ \vdots & \vdots & \vdots \\ 1 & x_n & \sin(2\pi x_n/4) \end{bmatrix}$$

(ii) (Highly Optional) GDP data for US economy is available at http://www.bea.gov/. Using the above, can you find the GDP growth trend of the US economy.

This is left as an exercise.